

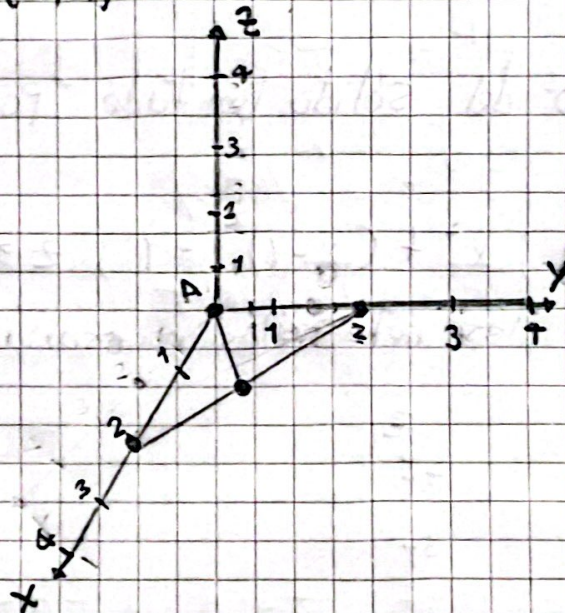
① El área de la región R_{xy} viene dada por

$$\int_0^1 \int_0^y dx dy + \int_1^2 \int_0^{2-y} dx dy$$

o dibule la región R_{xy} y calcule la integral en el orden $dy dx$.

$$\iint_R dA = \int_{y=0}^1 \int_{x=0}^{2-y} dx dy + \int_{y=1}^2 \int_{x=0}^y dx dy$$

Región R : límites por $X=0$, $X=2-y$ (para $0 \leq y \leq 1$)
y $X=y$ (para $1 \leq y \leq 2$). Intersección de $X=2-y$
 $X=y$ en $(1,1)$



puntos $A(0,0)$, $B(2,0)$, $C(1,1)$, $D(0,2)$. Región =
triángulo ABC o triángulo ACD

si $0 \leq x \leq 1$, y va de 0 a 2

si $1 \leq x \leq 2$, y toma los intervalos $0 \leq y \leq 2-x$

y $x \leq y \leq 2$.

$$\iint_R dA = \int_{x=0}^1 \left(\int_{y=0}^2 dy \right) dx + \int_{x=1}^2 \left(\int_{y=0}^{2-x} dy + \int_{y=x}^2 dy \right) dx$$

$$\int_0^1 \left(\int_0^2 dy \right) dx = \int_0^1 2 dx = 2$$

$$\int_1^2 \left(\int_0^{2-x} dy + \int_x^2 dy \right) dx = \int_1^2 ((2-x) + (2-x)) dx = \int_1^2 2(2-x) dx$$

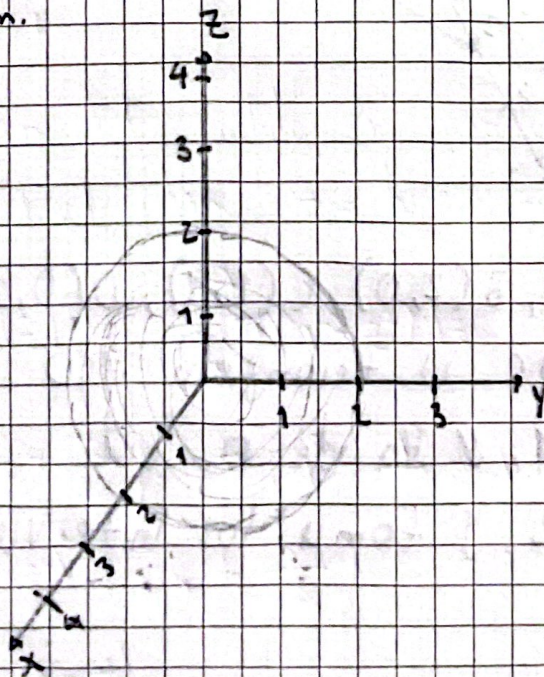
$$\int_1^2 2(2-x) dx = 2 \left[2x - \frac{x^2}{2} \right]_1^2 = 2 \left((4-2) - 2\left(2-\frac{1}{2}\right) \right) = 2(0.5) = 1$$

$$\text{Área}(R) = 2 + 1 = \underline{\underline{3}}$$

② Elabore el dibujo del sólido limitado por la Superficies

$$x^2 + y^2 + z^2 = 4; \quad x^2 + (y-1)^2 = 1; \quad z \geq 0$$

Plantear y Solucionar las integrales necesarias para Calcular el Volumen.



$$V = \int_{y=0}^1 \int_{x=0}^{\sqrt{4-y^2}} \int_{z=0}^{2-y} dz dx dy$$

$$\int_0^{2-y} dz = 2-y$$

$$V = \int_0^1 \int_0^{\sqrt{4-y^2}} (2-y) dx dy = \int_0^1 (2-y) \sqrt{4-y^2} dy$$

$$V = 2 \int_0^1 \sqrt{4-y^2} dy - \int_0^1 y \sqrt{4-y^2} dy$$

$$(a) I_1 = \int_0^1 \sqrt{4-y^2} dy$$

$$\int \sqrt{4-y^2} dy = \frac{y}{2} \sqrt{4-y^2} + 2 \arcsen\left(\frac{y}{2}\right) + C$$

$$I_1 = \frac{\sqrt{3}}{2} + \frac{\pi}{3}$$

$$(b) I_2 = \int_0^1 y \sqrt{4-y^2} dy$$

Sustitucion $u = 4 - y^2 \Rightarrow du = -2y dy$

$$I_2 = -\frac{1}{2} \int_4^3 u^{1/2} du = \frac{1}{2} \int_3^4 u^{1/2} du = \frac{1}{3} (4^{3/2} - 3^{3/2})$$

$$V = \frac{2\pi}{3} + 2\sqrt{3} - \frac{8}{3} \approx 2,89 = \frac{8}{3} - \sqrt{3}$$

④ Sea Q el sólido limitado por

$$z = 1; z = x^2 + y^2; z = 4$$

Calcular

$$\iiint_Q \frac{1}{\sqrt{x^2 + y^2 + 1}} dv$$

$$1 \leq z \leq 4 \quad y \quad x^2 + y^2 \leq z$$

$$I = \iiint_Q \frac{1}{\sqrt{x^2 + y^2 + 1}} dv$$

$$I = \int_{z=1}^4 \int_{\theta=0}^{2\pi} \int_{r=0}^{\sqrt{z}} \frac{1}{\sqrt{r^2 + 1}} r dr d\theta dz$$

$$I = 2\pi \int_1^4 \left(\int_0^{\sqrt{z}} \frac{r}{\sqrt{r^2 + 1}} dr \right) dz$$

$$\int_0^{\sqrt{z}} \frac{r}{\sqrt{r^2 + 1}} dr = \frac{1}{2} \int_{u=1}^{u=z+1} u^{-1/2} du = \frac{1}{2} \cdot 2 (u^{1/2}) \Big|_1^{z+1}$$

$$I = 2\pi \int_1^4 (\sqrt{z+1} - 1) dz = \sqrt{z+1} - 1$$

$$I = 2\pi \left(\int_1^4 \sqrt{z+1} dz - \int_1^4 1 dz \right)$$

$$\int_1^4 \sqrt{z+1} dz = \int_2^5 t^{1/2} dt = \frac{2}{3} t^{3/2} \Big|_2^5 = \frac{2}{3} (5^{3/2} - 2^{3/2})$$

$$I = 2\pi \left(\frac{2}{3} (5^{3/2} - 2^{3/2}) - 3 \right)$$

$$I = \frac{4\pi}{3} (5\sqrt{5} - 2\sqrt{2}) - 6\pi$$

$$I \approx 16,11$$

5) Calcular el Volumen del sólido Q limitado por $x^2 + y^2 + z^2 = a^2$ y el plano $z = h$; $0 < h < a$

$$V = \int_{z=h}^a A(z) dz = \int_h^a \pi(a^2 - z^2) dz$$

$$V = \pi \left[a^2 z - \frac{z^3}{3} \right]_h^a = \pi \left(a^3 - \frac{a^3}{3} - a^2 h + \frac{h^3}{3} \right)$$

$$V = \pi \left(\frac{2a^3}{3} - a^2 h + \frac{h^3}{3} \right)$$

$$V = \frac{\pi H^2 (3a - H)}{3}$$

$$③ \quad V = \iint_R \sqrt{4 - (x^2 + y^2)} dx dy$$

coordenadas
Polares

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dx dy = r dr d\theta$$

$$r^2 - 2r \sin \theta + 1 = 1 \Rightarrow r = 2 \sin \theta$$

$$0 \leq r \leq 2 \sin \theta, \quad 0 \leq \theta \leq \pi$$

$$V = \int_0^\pi \int_0^{2 \sin \theta} \sqrt{4 - r^2} r dr d\theta$$

$$\text{Sea } u = 4 - r^2 \Rightarrow du = -2r dr \Rightarrow r dr = -\frac{1}{2} du$$

$$\int r \sqrt{4 - r^2} dr = -\frac{1}{2} \int u^{1/2} du = -\frac{1}{3} u^{3/2} + C$$

$$\int_0^{2 \sin \theta} r \sqrt{4 - r^2} dr = \frac{1}{3} \left[0 - (4 - 4 \sin^2 \theta)^{3/2} \right] = \frac{8}{3} (1 - |\cos^3 \theta|)$$

$$V = \frac{8}{3} \int_0^\pi (1 - |\cos^3 \theta|) d\theta = \frac{8}{3} \left[\int_0^{\pi/2} (1 - \cos^3 \theta) d\theta + \int_{\pi/2}^\pi (1 + \cos^3 \theta) d\theta \right]$$

$$\int_0^{\pi/2} \cos^3 \theta d\theta = \frac{2}{3}, \quad \int_{\pi/2}^\pi \cos^3 \theta d\theta = -\frac{2}{3}$$

$$V = \frac{8}{3} \left[\left(\frac{\pi}{2} - \frac{2}{3} \right) + \left(\frac{\pi}{2} - \frac{2}{3} \right) \right] = \frac{8}{3} \left(\pi - \frac{4}{3} \right) = \frac{8\pi}{3} - \frac{32}{9}$$

$$V = \frac{8\pi}{3} - \frac{32}{9} //$$

